

Internet of Things (Section C)⇒

Real-Time Edge Analytics & Data Publishing →

Summery ⇒

In this phase the project transitions from offline pre stored data analysis to Real-Time Edge analytics system , through the exisiting model from Part B (which has properties which were built on mobility , Intensity and direction change scores) we have built an live sensor data streams from mobile through the matlab mobile application.

through this live data from mobile the system we get an immidiate feedback to users about their active(running/walking) and postural(sitting/standing) states in real time

System Details ⇒

1. Edge device (mobile) →

captures the tri-axial acceleration and orientation data from the tasks performed

2. processing →

from the raw data we perform real time signal cleaning through butterworth and Savitzky-golay filters and then we extract the features to classify the data

3. Cloud (MATLAB) →

And then finally through the cloud we intend to publish the data continuously for long term trackings.

Algorithm Explanation ⇒

Butterworth low pass filter : A 4th-order filter with a 5 Hz cutoff is used to remove high-frequency noise, as human movement typically occurs at lower frequencies

Savitzky-Golay Smoothing: This preserves the essential "peaks" of physical actions (like footsteps) while smoothing out the remaining signal ripples

Coordinate Independence: The system calculates the Euclidean Norm (Magnitude) of the acceleration to ensure detection remains accurate regardless of how the phone is oriented in a pocket

The system calculates three core values of motion in real-time:

Mobility Score: Measures the variance $\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$ of the filtered acceleration. High variance indicates erratic movement like running.

Intensity Score: Calculated as the Mean Square of the signal to measure the raw physical "power" or force behind an activity.

Direction Change: Tracks rotation by summing absolute differences in orientation. This helps distinguish between steady movement and frequent turning.

Classification and scoring :

The **Current Activity Score (CAS)** ; It normalizes all features to a [0, 1] range to prevent outliers from dominating the score

$$CAS = (0.4 \cdot mob_n) + (0.4 \cdot int_n) + (0.2 \cdot dir_n)$$

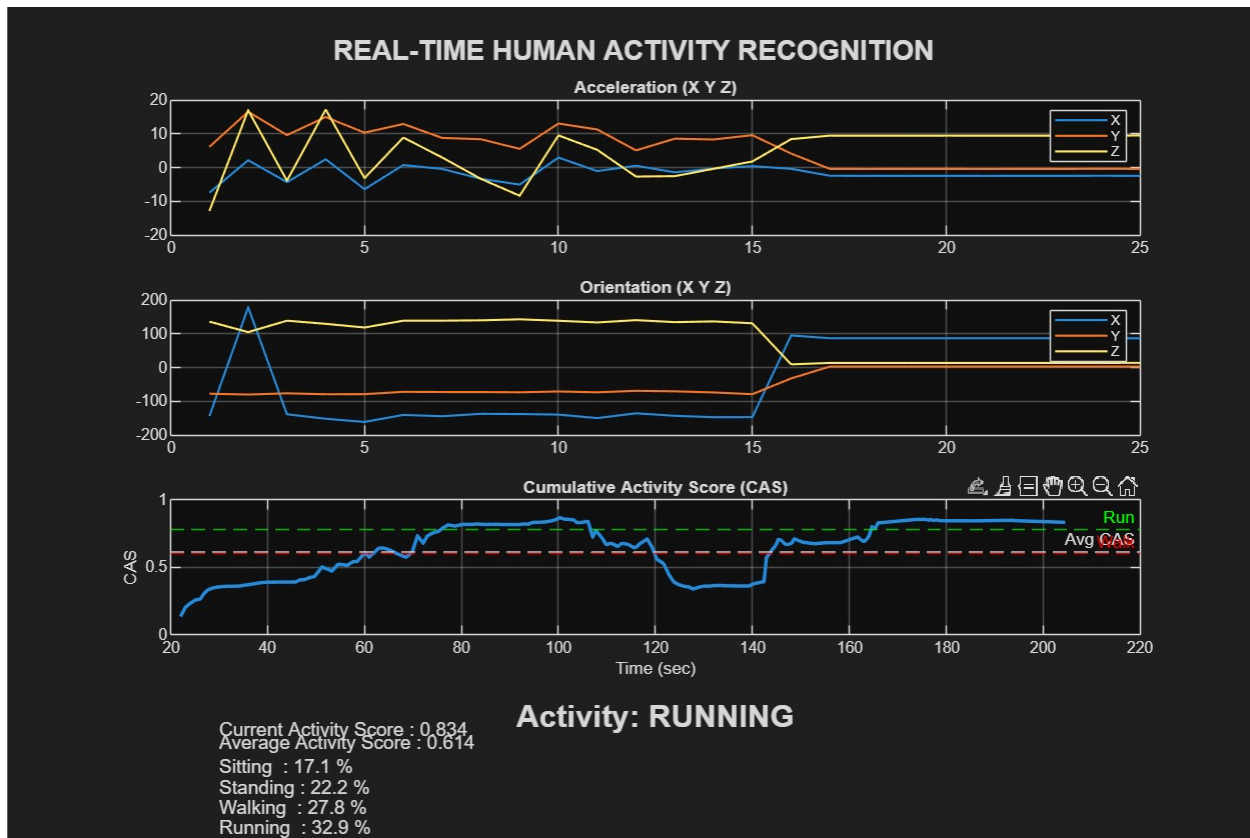
Average Activity score (AAS) : the average activity score is nothing but the average of all the current activity scores till the time stamp and this would help in estimating the average active time of a person and this would help in estimating the average active period and how active a person is .

at the end of each second the system prints out the output along with an instant score and average activity score & also at the end it results out the percentage of walking , running , sitting and standing throughout the process of testing .

Activity CAS range list →

Activity	CAS Range	Orientation Condition
SITTING	< 0.60	Low (<15 degrees)
STANDING	< 0.60	High (>45 degrees)
WALKING	0.60 to 0.78	N/A

RUNNING	> 0.78	N/A
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Here the the functional MATLAB GUI is capturing live movement data

The top two subplots display real-time **Acceleration** and **Orientation** waveforms, showing the raw physical inputs from the smartphone.

The third plot tracks the **Current Activity Score (CAS)** over time. In this specific snapshot, the CAS line is above the green dashed threshold (0.78), indicating high-intensity movement.

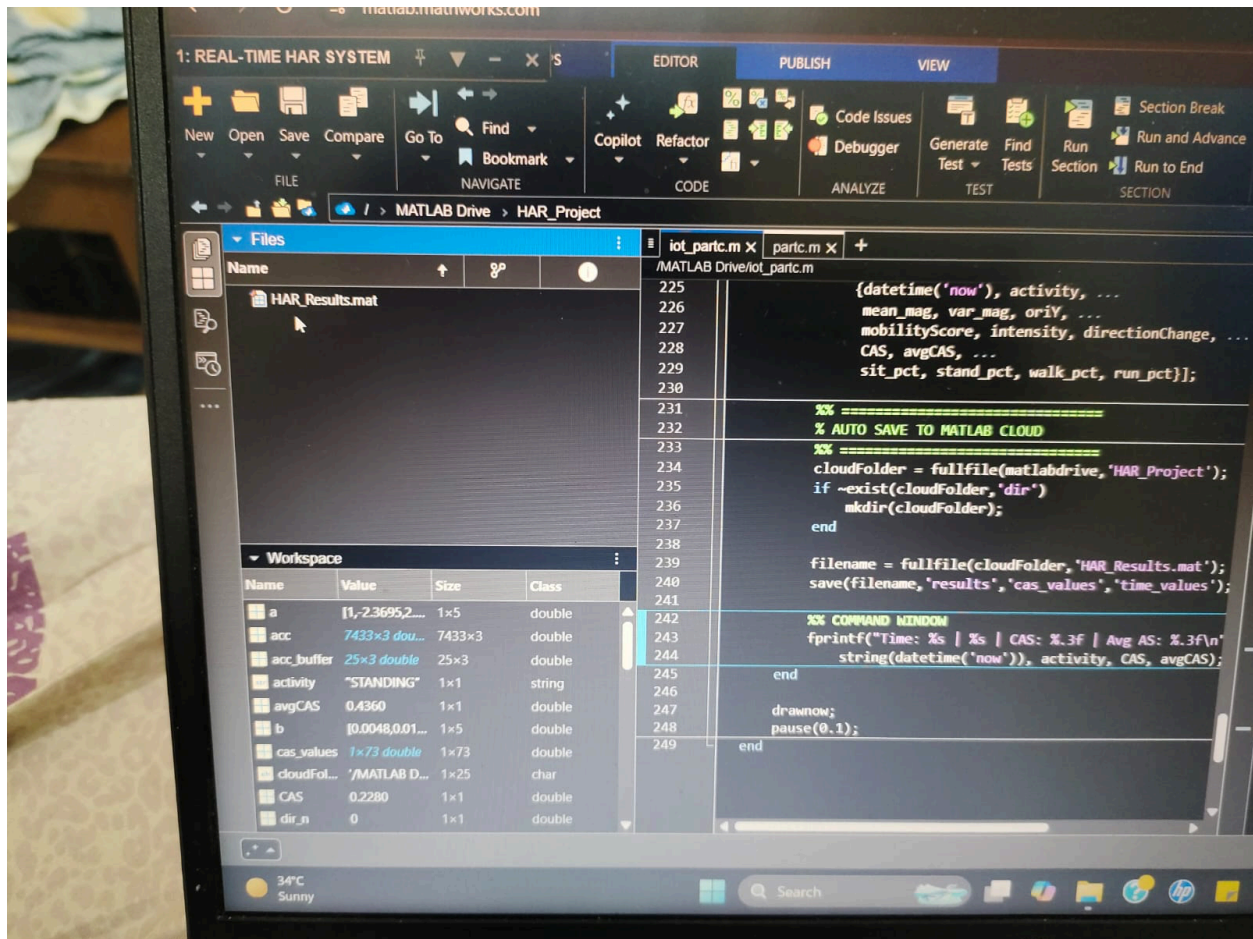
The bottom panel clearly identifies the current activity as **"RUNNING"** and provides a percentage breakdown of the entire session (e.g., Running: 32.9%, Walking: 27.8%).

Time: 25-Apr-2026 12:18:09	STANDING	CAS: 0.575	AvgCAS: 0.426
Time: 25-Apr-2026 12:18:10	WALKING	CAS: 0.615	AvgCAS: 0.430
Time: 25-Apr-2026 12:18:11	WALKING	CAS: 0.733	AvgCAS: 0.436
Time: 25-Apr-2026 12:18:12	WALKING	CAS: 0.681	AvgCAS: 0.442
Time: 25-Apr-2026 12:18:13	WALKING	CAS: 0.733	AvgCAS: 0.448
Time: 25-Apr-2026 12:18:15	WALKING	CAS: 0.761	AvgCAS: 0.454
Time: 25-Apr-2026 12:18:15	WALKING	CAS: 0.766	AvgCAS: 0.460
Time: 25-Apr-2026 12:18:17	RUNNING	CAS: 0.797	AvgCAS: 0.467
Time: 25-Apr-2026 12:18:18	RUNNING	CAS: 0.816	AvgCAS: 0.474
Time: 25-Apr-2026 12:18:19	RUNNING	CAS: 0.806	AvgCAS: 0.480
Time: 25-Apr-2026 12:18:21	RUNNING	CAS: 0.820	AvgCAS: 0.486
Time: 25-Apr-2026 12:18:22	RUNNING	CAS: 0.819	AvgCAS: 0.492
Time: 25-Apr-2026 12:18:23	RUNNING	CAS: 0.819	AvgCAS: 0.498
Time: 25-Apr-2026 12:18:24	RUNNING	CAS: 0.822	AvgCAS: 0.504

the above image is confirmation of continuous data saving process and Each line represents a distinct processing window with a timestamp.

The log shows the system transitioning from **STANDING** (CAS: 0.575) to **WALKING** (CAS: 0.615) and finally to **RUNNING** (CAS: 0.797 and above) as the user's intensity increases.

finally It confirms that the classification logic matches the calculated CAS values in real-time.



"Edge-to-Cloud" implementation → file located within the **HAR_Project** folder on **MATLAB Drive**, and the processed data is being saved to the cloud as it is streamed.

Code ⇒

```
clc; clear; close all;

%% =====
% SETTINGS
%% =====
fs = 50;
window_size = 0.5;
maxSamples = fs * window_size;
```

```

[b,a] = butter(4,5/(fs/2),'low');

%% =====
% MOBILE CONNECTION
%% =====
m = mobiledev;

if ~m.Connected
    error('Mobile not connected! Open MATLAB Mobile.');
```

end

```

m.AccelerationSensorEnabled = 1;
m.OrientationSensorEnabled = 1;
m.Logging = 1;

disp("📶 Real-time system running...");

%% =====
% BUFFERS
%% =====
acc_buffer = [];
ori_buffer = [];
results = [];

%% =====
% FIGURE
%% =====
figure('Name','REAL-TIME HAR SYSTEM', ...
       'Position',[100 100 1000 750]);

%% =====
% VARIABLES
%% =====
prevCAS = 0;

```

```

% CAS plot storage
cas_values = [];
time_values = [];

% Activity counters
sit_time = 0;
stand_time = 0;
walk_time = 0;
run_time = 0;

startTime = tic;

%% =====
% MAIN LOOP
%% =====
while true

    %% SENSOR DATA
    [acc,~] = accellog(m);
    [ori,~] = orientlog(m);

    if isempty(acc) || isempty(ori)
        pause(0.05);
        continue;
    end

    acc_buffer = [acc_buffer; acc(end,:)];
    ori_buffer = [ori_buffer; ori(end,:)];

    % Sliding window
    if size(acc_buffer,1) > maxSamples
        acc_buffer(1,:) = [];
        ori_buffer(1,:) = [];
    end

    %% =====

```



```

% PLOT 1: ACCELERATION
%% =====
subplot(4,1,1);
plot(acc_buffer,'LineWidth',1.2);
title('Acceleration (X Y Z)');
legend('X','Y','Z');
grid on;

%% =====
% PLOT 2: ORIENTATION
%% =====
subplot(4,1,2);
plot(ori_buffer,'LineWidth',1.2);
title('Orientation (X Y Z)');
legend('X','Y','Z');
grid on;

%% =====
% PROCESSING
%% =====
if size(acc_buffer,1) > 20

    %% Magnitude
    mag = sqrt(sum(acc_buffer.^2,2));
    mag = filtfilt(b,a,mag);
    mag = sgolayfilt(mag,3,11);

    mean_mag = mean(mag);
    var_mag   = var(mag);

    %% Orientation
    oriY = ori_buffer(end,2);
    oriX = ori_buffer(:,1);

    %% Scores
    mobilityScore = var_mag;

```



```

intensity      = mean(mag.^2);
directionChange = sum(abs(diff(oriX)));

%% Normalize
mob_n = mobilityScore/(mobilityScore+1);
int_n = intensity/(intensity+50);
dir_n = directionChange/(directionChange+100);

%% CAS
CAS = 0.4*mob_n + 0.4*int_n + 0.2*dir_n;

%% Smooth CAS
CAS = 0.5*prevCAS + 0.5*CAS;
prevCAS = CAS;

%% =====
% CLASSIFICATION
%% =====
if CAS < 0.600 && abs(oriY)<15
    activity = "SITTING";

elseif CAS < 0.600 && abs(oriY)>45
    activity = "STANDING";

elseif CAS >= 0.600 && CAS < 0.780
    activity = "WALKING";

else
    activity = "RUNNING";
end

%% =====
% COUNTERS
%% =====
dt = 0.1;

```

```

switch activity
    case "SITTING"
        sit_time = sit_time + dt;
    case "STANDING"
        stand_time = stand_time + dt;
    case "WALKING"
        walk_time = walk_time + dt;
    case "RUNNING"
        run_time = run_time + dt;
end

totalTime = sit_time + stand_time + walk_time + run_time;

%% Percentages
sit_pct    = (sit_time    / totalTime) * 100;
stand_pct  = (stand_time  / totalTime) * 100;
walk_pct   = (walk_time   / totalTime) * 100;
run_pct    = (run_time    / totalTime) * 100;

%% =====
% STORE CAS
%% =====
elapsed = toc(startTime);

time_values(end+1) = elapsed;
cas_values(end+1)  = CAS;

avgCAS = mean(cas_values);

%% =====
% PLOT 3: CAS GRAPH
%% =====
subplot(4,1,3);
plot(time_values,cas_values,'LineWidth',2);
hold on;

```

```

    yline(avgCAS,'--','Avg CAS','LineWidth',1.2);
    yline(0.60,'--r','Walk');
    yline(0.78,'--g','Run');
    hold off;
    grid on;
    title('Cumulative Activity Score (CAS)');
    xlabel('Time (sec)');
    ylabel('CAS');

    %% =====
    % DISPLAY PANEL
    %% =====
    subplot(4,1,4);
    cla;

    text(0.50,0.88,"Activity: "+activity,...
        'FontSize',18,...
        'FontWeight','bold',...
        'HorizontalAlignment','center');

    text(0.05,0.78,sprintf("Current Activity Score : %.3
f",CAS),'FontSize',11);
    text(0.05,0.68,sprintf("Average Activity Score : %.3
f",avgCAS),'FontSize',11);

    text(0.05,0.50,sprintf("Sitting   : %.1f %%",sit_pc
t),'FontSize',11);
    text(0.05,0.35,sprintf("Standing  : %.1f %%",stand_pc
t),'FontSize',11);
    text(0.05,0.20,sprintf("Walking   : %.1f %%",walk_pc
t),'FontSize',11);
    text(0.05,0.05,sprintf("Running   : %.1f %%",run_pc
t),'FontSize',11);

    axis off;

```

```

sgtitle("REAL-TIME HUMAN ACTIVITY RECOGNITION", ...
        'FontSize',16,'FontWeight','bold');

%% =====
% SAVE RESULTS TO MEMORY
%% =====
results = [results;
           {datetime('now'), activity, ...
            mean_mag, var_mag, oriY, ...
            mobilityScore, intensity, directionChange, ...
            CAS, avgCAS, ...
            sit_pct, stand_pct, walk_pct, run_pct}];

%% =====
% AUTO SAVE TO MATLAB CLOUD
%% =====
cloudFolder = fullfile(matlabdrive,'HAR_Project');
if ~exist(cloudFolder,'dir')
    mkdir(cloudFolder);
end

filename = fullfile(cloudFolder,'HAR_Results.mat');
save(filename,'results','cas_values','time_values');

%% COMMAND WINDOW
fprintf("Time: %s | %s | CAS: %.3f | AvgCAS: %.3f\n",
...
        string(datetime('now')), activity, CAS, avgCAS);
end

drawnow;
pause(0.1);
end

```

Conclusion :

- **Sensor Activation:** The system automatically enables the `AccelerationSensorEnabled` and `OrientationSensorEnabled` on the connected mobile device to begin data streaming.
- **Real-Time Data Acquisition:** It continuously acquires 3-axis acceleration and orientation data, storing them in sliding buffers to maintain a 0.5-second window of the most recent samples.
- **Filtering and Smoothing:** A 4th-order Butterworth low-pass filter and a Savitzky-Golay filter are applied to the raw data to remove high-frequency noise and jitter.
- **Feature Extraction:** Required values such as mobility score, intensity, and direction change are calculated from the filtered sensor magnitude and orientation.
- **CAS and AAS Calculation:** The system calculates the Current Activity Score (CAS) by fusing the extracted features and maintains a running Average Activity Score (AAS).
- **Graphical Plotting:** Real-time plots are generated for raw acceleration, orientation, and the CAS values.
- **Activity Percentage Tracking:** The system monitors the duration of each classified state—sitting, standing, walking, or running—and calculates the cumulative percentage of time spent in each activity.
- **Cloud Data Storage:** All processed results and parameters are automatically saved to a `.mat` file in the MATLAB Drive cloud folder for persistent storage.